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Properties of Materials
Lecture 2

Dr. Alkis Tsamis
at441@leicester.ac.uk

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Lecture 1: Module Overview, Module Aim, Why Study Materials, The World of Materials (1), History - The Evolution of Materials, Materials Selection in Design (1), Properties of Engineering Materials (1), Products, Three-Way Relationship (1), Origins of Material Properties, How Do We - As Engineers - Select Materials
Lecture 2: Concepts and Terminology (1), Properties of Engineering Materials (2), Mechanical Properties (1), Elastic Properties, Elastic Properties: Covalent and Metallic Bonds - Ionic Model - The Origin of Modulus, Plastic Properties, Fracture / Fatigue / Creep, Thermal Properties (1), Density and Corrosion, Corrosion: Anodic and Cathodic Reactions, Ellingham Diagram
Lecture 3: Structure-Property-Processing Relationship, Structure Sensitive Properties, Structure Insensitive Properties, The Three-Part Relationship, Materials Property Web Sites, Materials Selection, Material Property - Charts: Modulus - Density, Mechanical Properties (2), Thermal Properties (2), A TOOL for Materials Selection, Structure Sensitivity, Structure Sensitivity in Metals, Controlling Structure, Case Study

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Engineering Materials and Their Properties

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Summary

- The World of Materials is vast:
  - >150,000 different materials created or discovered to date
- As Engineers, we need to be able to:
  - **Select the optimum material** for any given application
  - Understand the factors that influence the properties and behaviour of materials – **strengthening mechanisms**
  - Predict and prevent common **failure mechanisms** from occurring
- With any luck*, in 5 weeks you'll be able to do just that!

*By 'luck', I of course mean studious learning, practice and paying attention in my sessions!

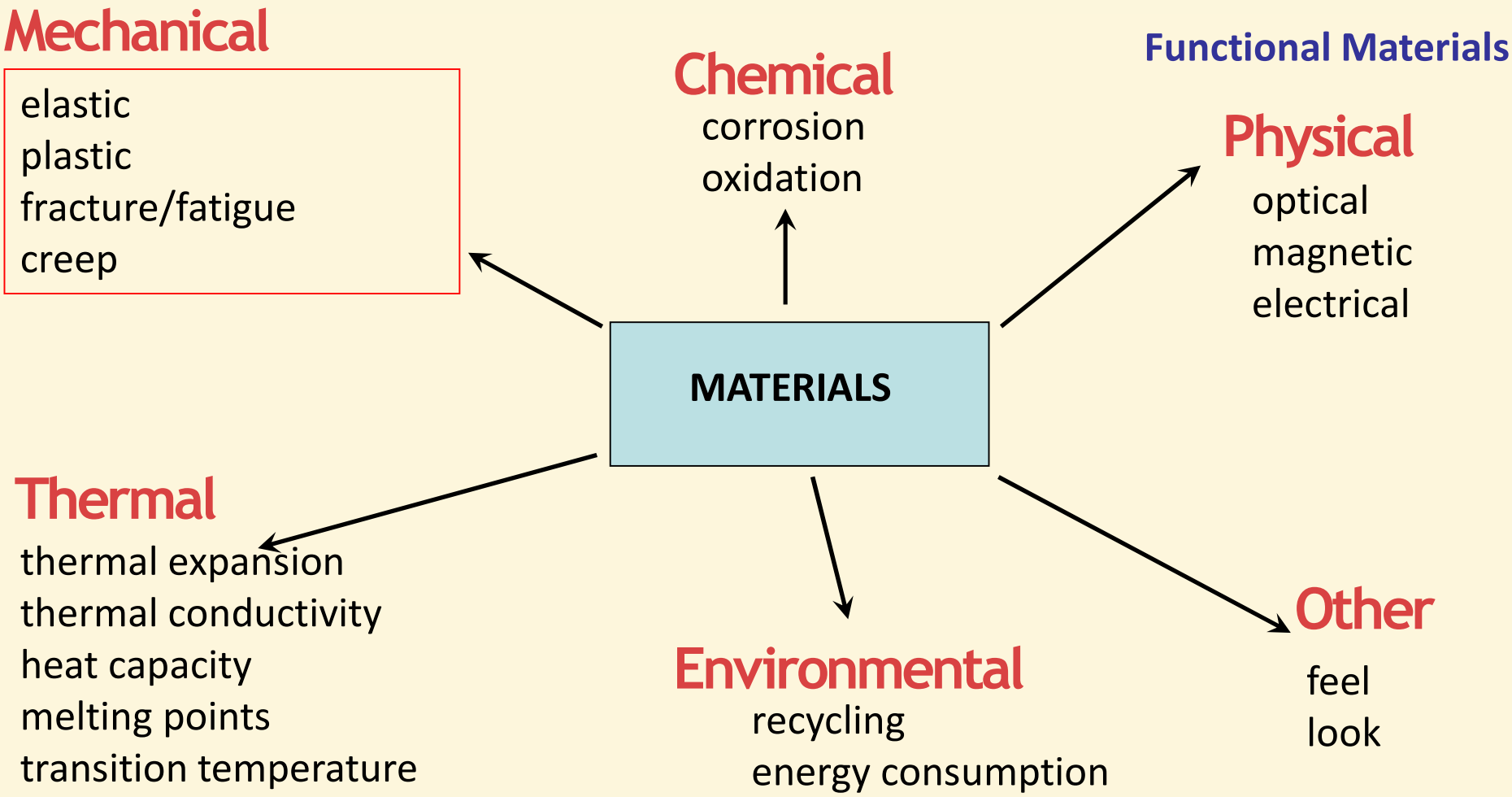
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Concepts and terminology

- Material **families**:
  - Metals, alloys
  - Polymers
  - Elastomers
  - Glasses
  - Ceramics
- (Some) material **properties**:
  - Stiffness
  - Strength
  - Toughness
- (Some) failure mechanisms:
  - Yielding
  - Creep
  - Fracture
  - Fatigue
  - Corrosion
- Three-way relationship:
  - Processing
  - Structure
  - Properties

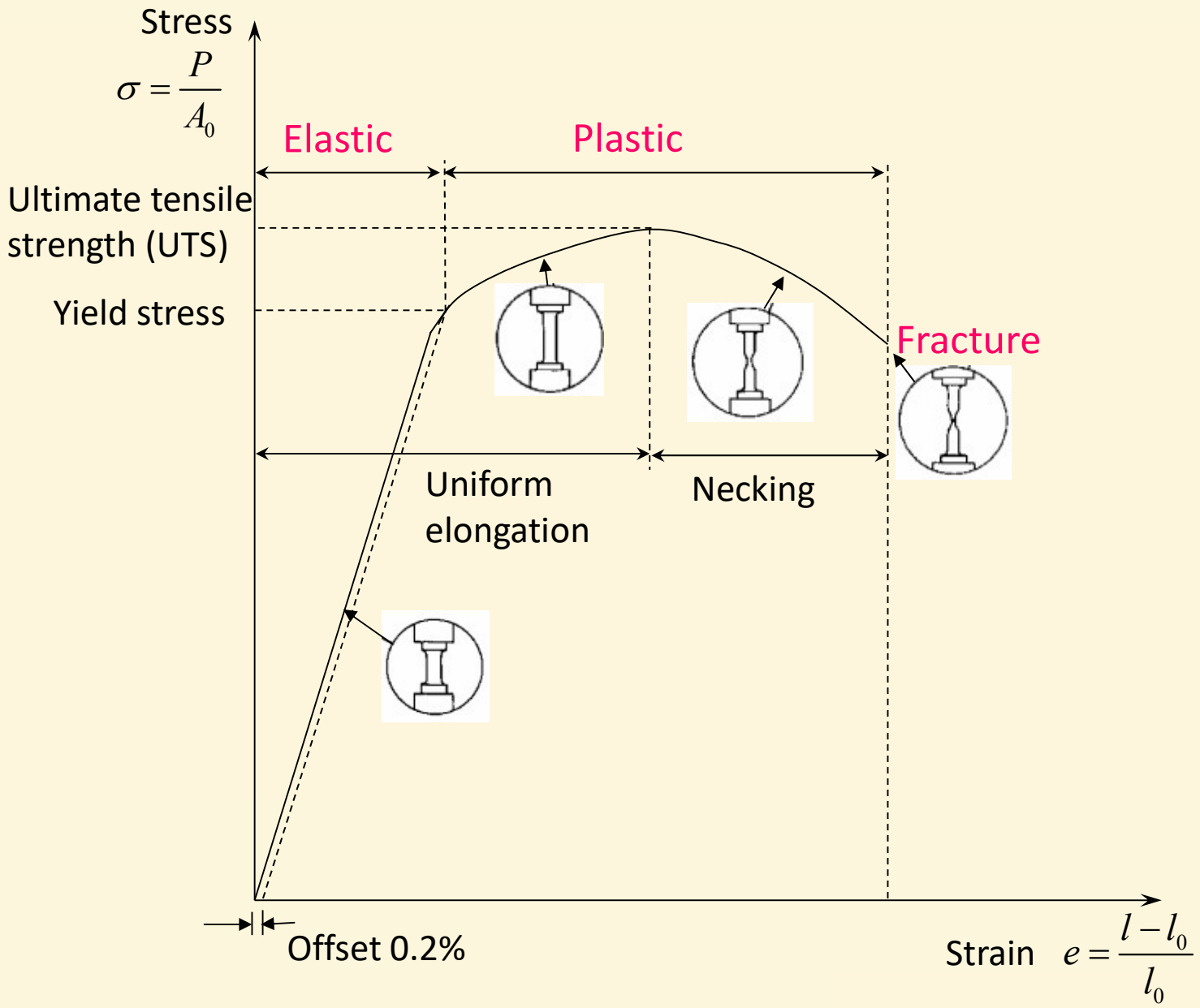
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Properties of Engineering Materials



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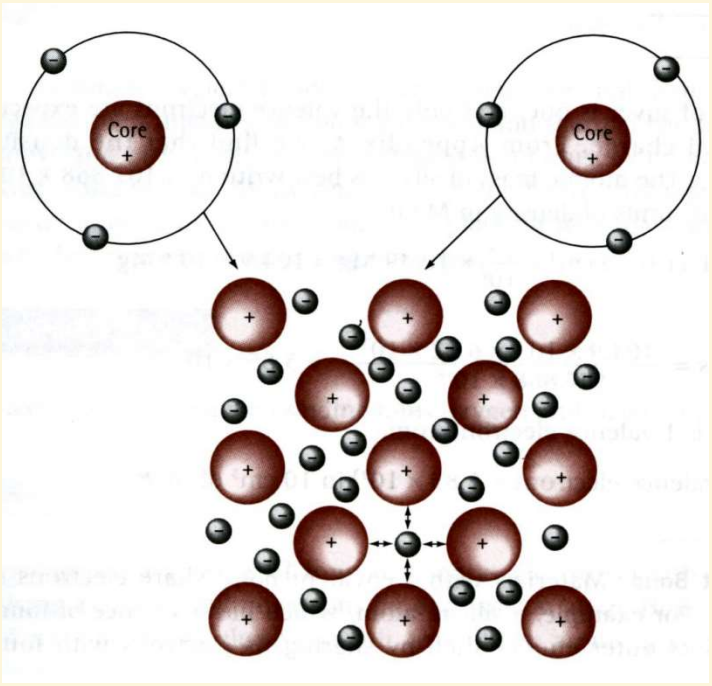
Review: Mechanical Properties



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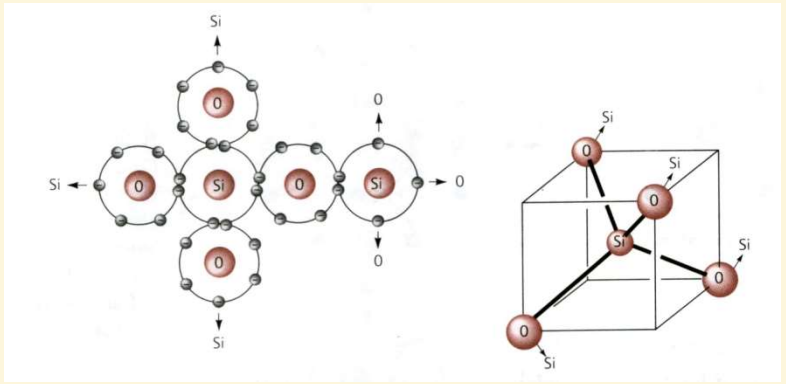
Elastic Properties

Modulus of elasticity (Young's Modulus E): slope of the stress-strain curve, determined by the strength (stiffness) of the atomic bonds and the nature of the atomic packing



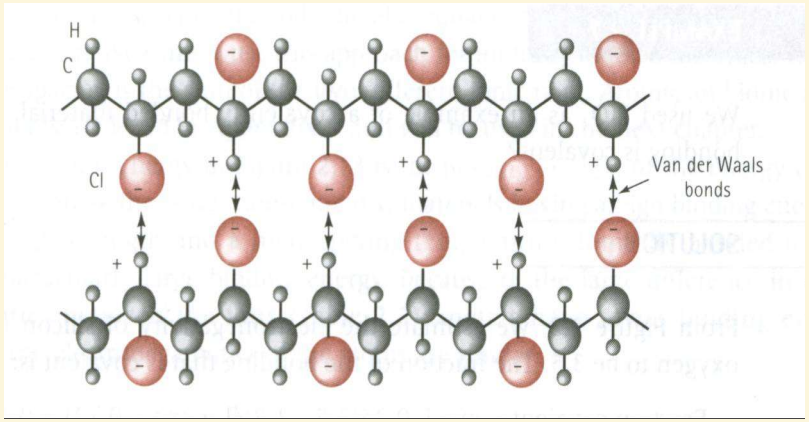
Metallic bond

Steel: 200GPa , Al: 70GPa



Covalent bonds

Ceramics: Al_2O_3 380GPa ($GN.m^{-2}$)



Van der Waals bonds

Polymers: Nylon 2-4 GPa

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Elastic Properties

Modulus of elasticity (Young's Modulus E): slope of the stress-strain curve, determined by the strength (stiffness) of the atomic bonds and the nature of the atomic packing. Some materials show non linear elastic behaviour….

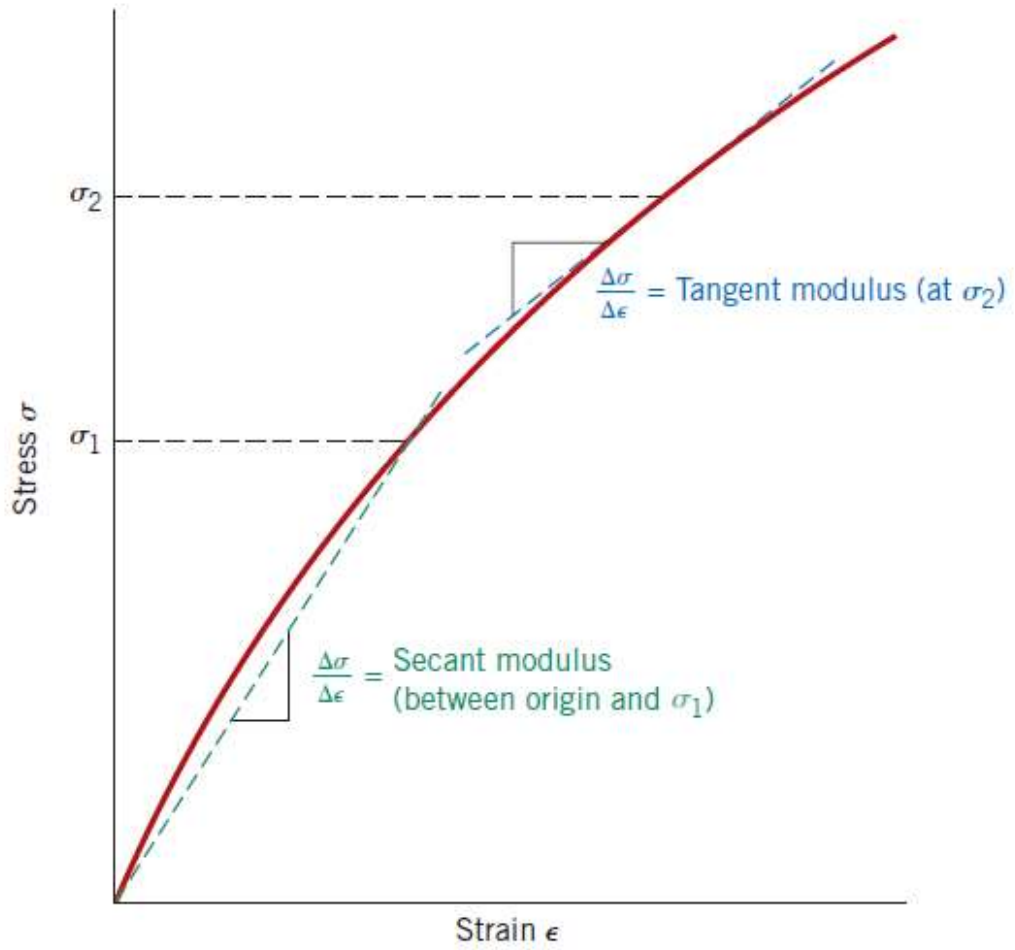


Figure 6.6 Schematic stress–strain diagram showing non-linear elastic behavior, and how secant and tangent moduli are determined.

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Elastic Properties

- **Modulus of elasticity (Young's Modulus E):**

Table 6.1 **Room-Temperature Elastic and Shear Moduli, and Poisson's Ratio for Various Metal Alloys**

	Modulus of Elasticity		*Shear Modulus*		*Poisson's Ratio*
Metal Alloy	*GPa*	*10^6 psi*	*GPa*	*10^6 psi*	
Aluminum	69	10	25	3.6	0.33
Brass	97	14	37	5.4	0.34
Copper	110	16	46	6.7	0.34
Magnesium	45	6.5	17	2.5	0.29
Nickel	207	30	76	11.0	0.31
Steel	207	30	83	12.0	0.30
Titanium	107	15.5	45	6.5	0.34
Tungsten	407	59	160	23.2	0.28

Elastic Properties: Covalent and Metallic bonds

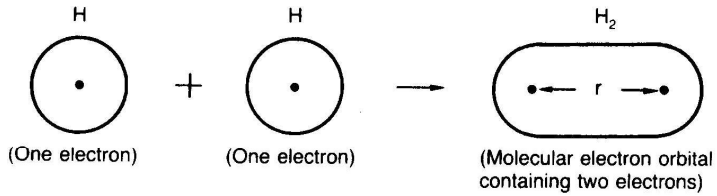
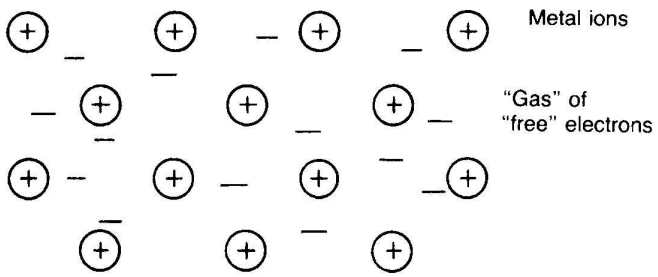
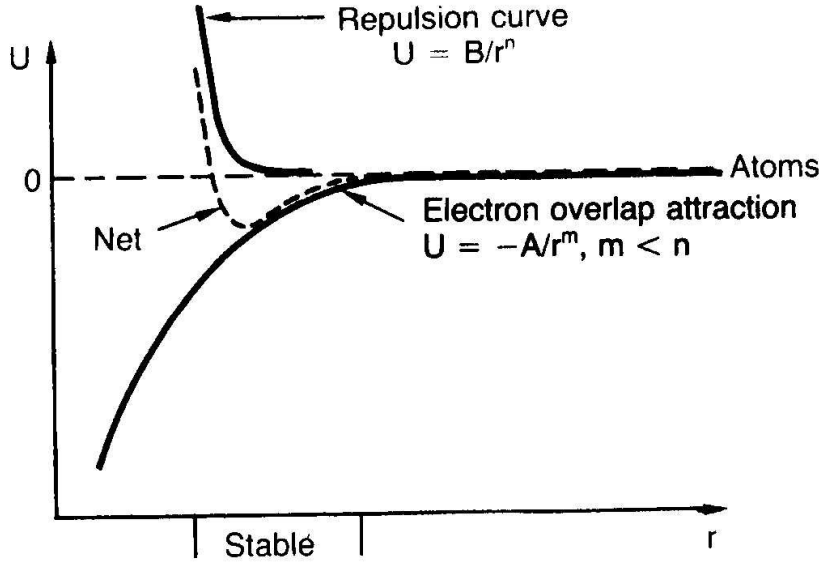


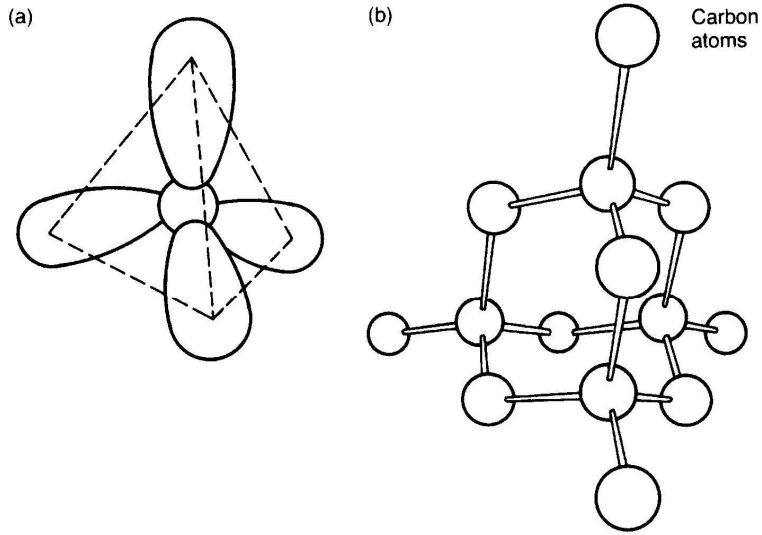
Fig. 4.5. The formation of a covalent bond – in this case between two hydrogen atoms, making a hydrogen *molecule*.



. Bonding in a metal – metallic bonding.



4.6. The formation of a covalent bond, viewed in terms of energy.



Directional covalent bonding in diamond.

Elastic Properties: Ionic Model

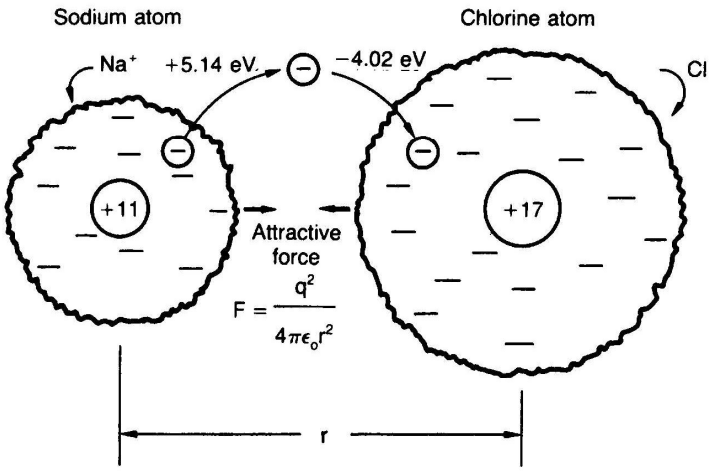


Fig. 4.3. The formation of an ionic bond – in this case between a sodium atom and a chlorine atom, making sodium chloride.

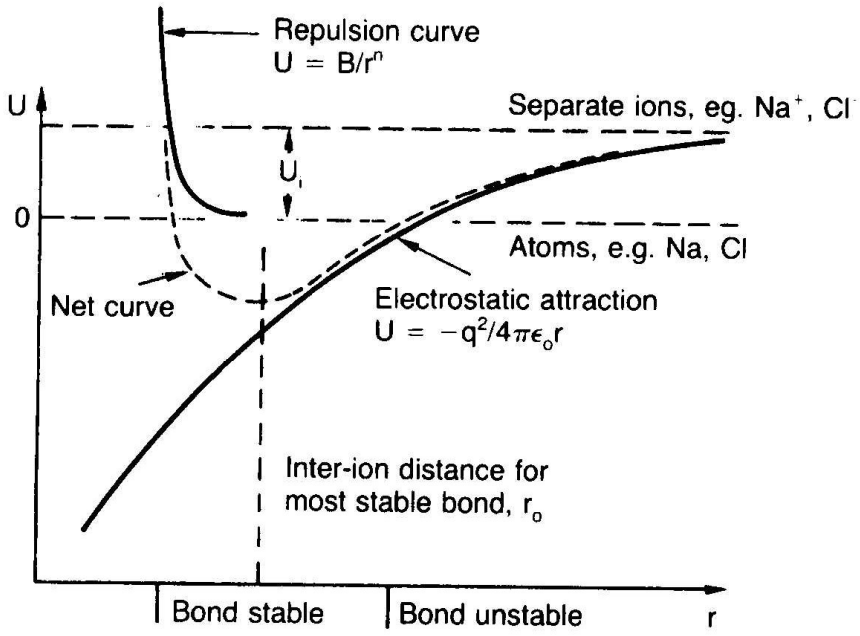
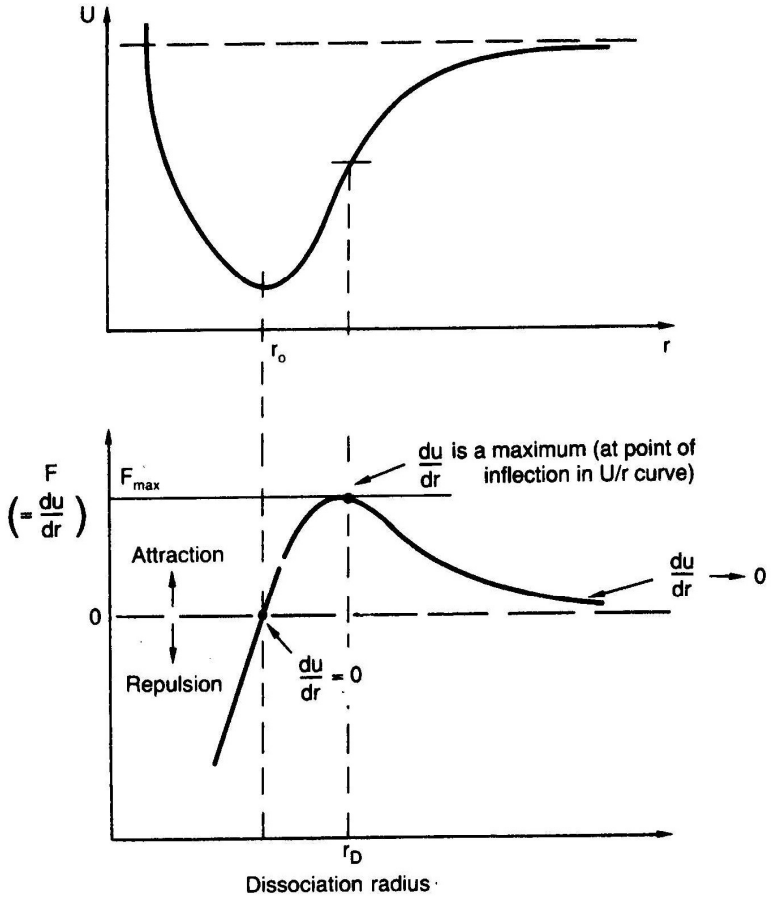
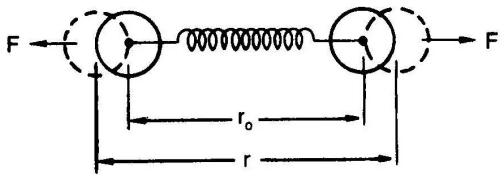


Fig. 4.4. The formation of ionic bond, viewed in terms of energy.

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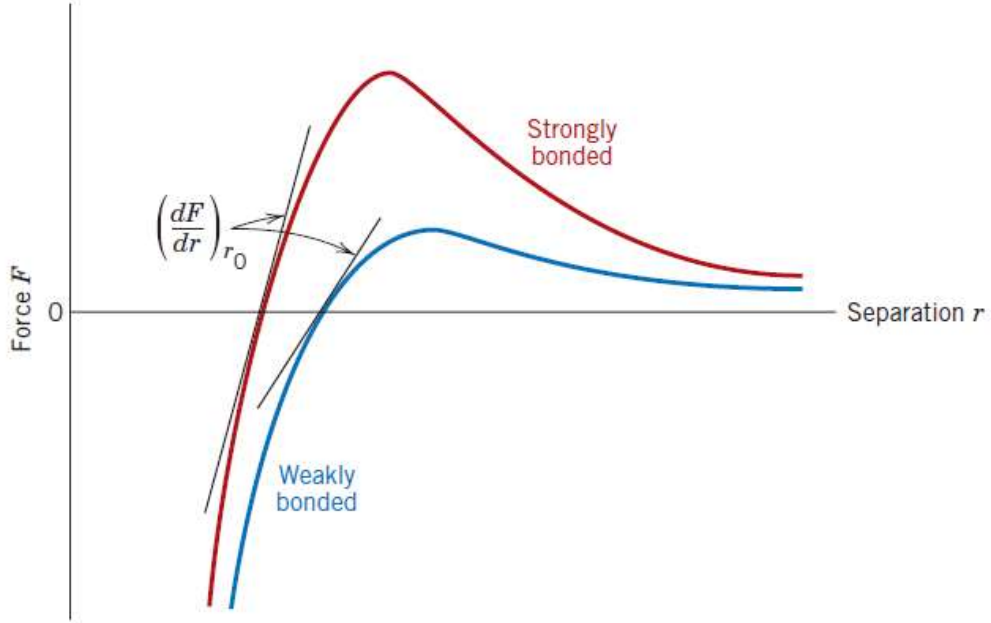
Elastic Properties: The Origin of Modulus





e energy curve (top), when differentiated (eqn. (4.6)) gives the force–distance curve (centre).

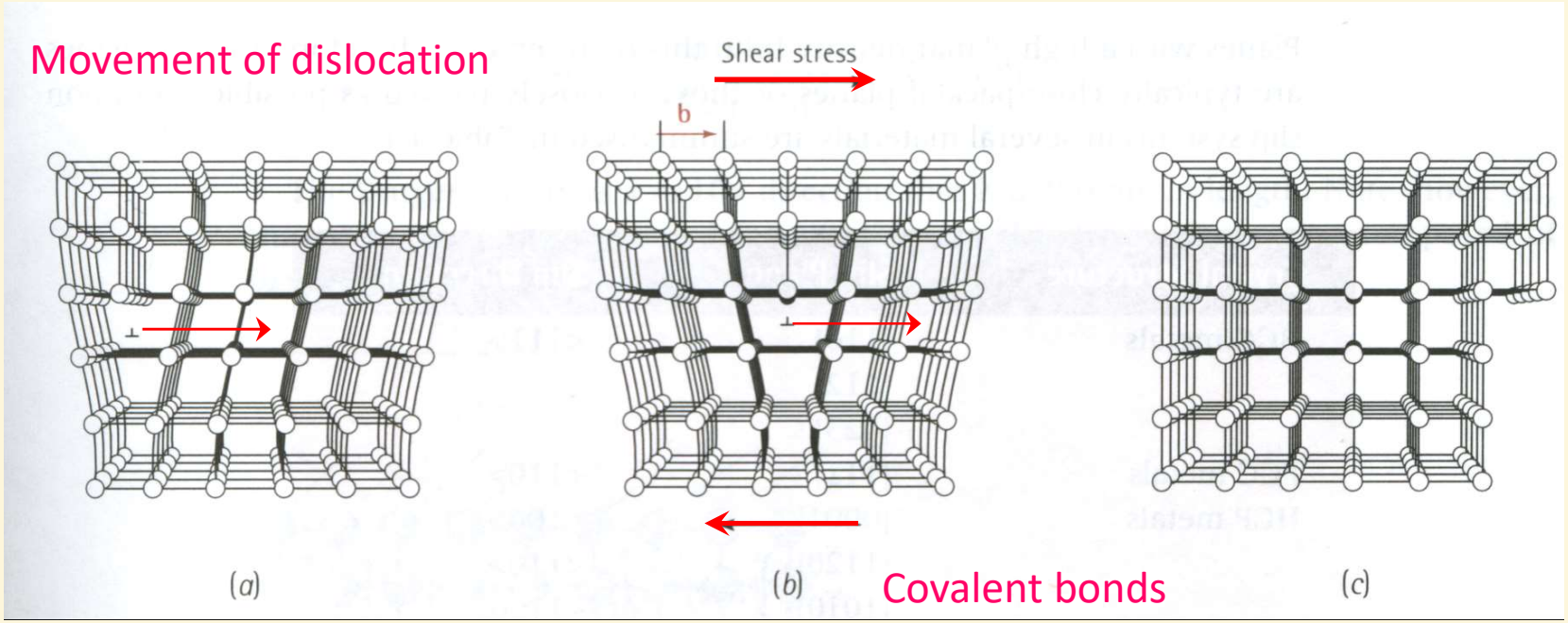
$$\frac{dF}{dr} = S$$



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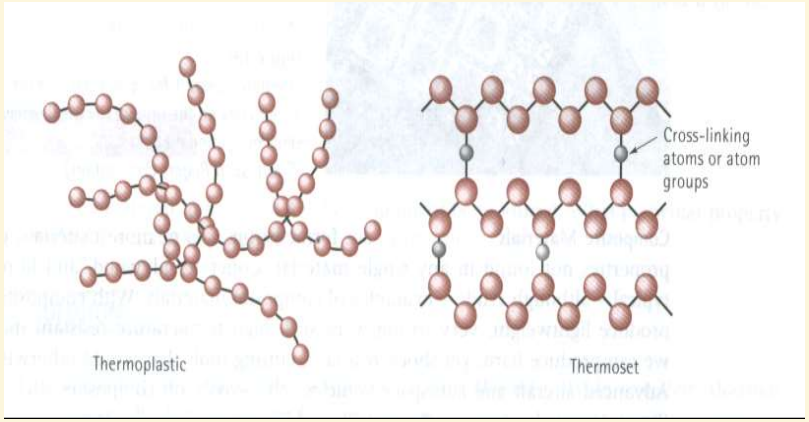
Plastic Properties

Plastic deformation: **dislocation movement in crystalline materials**



Slip systems in metals and ceramics

Crystal	# slip systems	Slip plane	Slip direction
BCC metals	48	{110}{112}{123}	<100>
FCC metals	12	{111}	<110>
HCP metals	3	{0001}{1120}…	<1120>
MgO, NaCl		{110}	<110>



Polymers : Thermoplastic & Thermosetting

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Fracture/Fatigue/Creep

Fracture: propagation of cracks through the body

usually increasing yield strength results in a decrease of toughness

mild steel: moderate yield strength – high toughness

ceramics: high yield strength – low toughness

Fatigue: failure due to a large number of applications of stress (cyclic stress below the yield strength)

Problems in metals and polymers, but less of a problem in ceramics

Creep: plastic deformation at high temperature

melting temperature T_m

metals: creep above $0.3T_m$ (superalloys can reach over $0.7T_m$)

ceramics: creep over $0.5T_m$

polymers: creep over T_g (glass transition temperature) and many polymers creep at room temperatures

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Thermal Properties

- Thermal Expansion

$$\frac{l_f - l_0}{l_0} = \alpha_l (T_f - T_0)$$

$$\frac{\Delta l}{l_0} = \alpha_l \Delta T$$

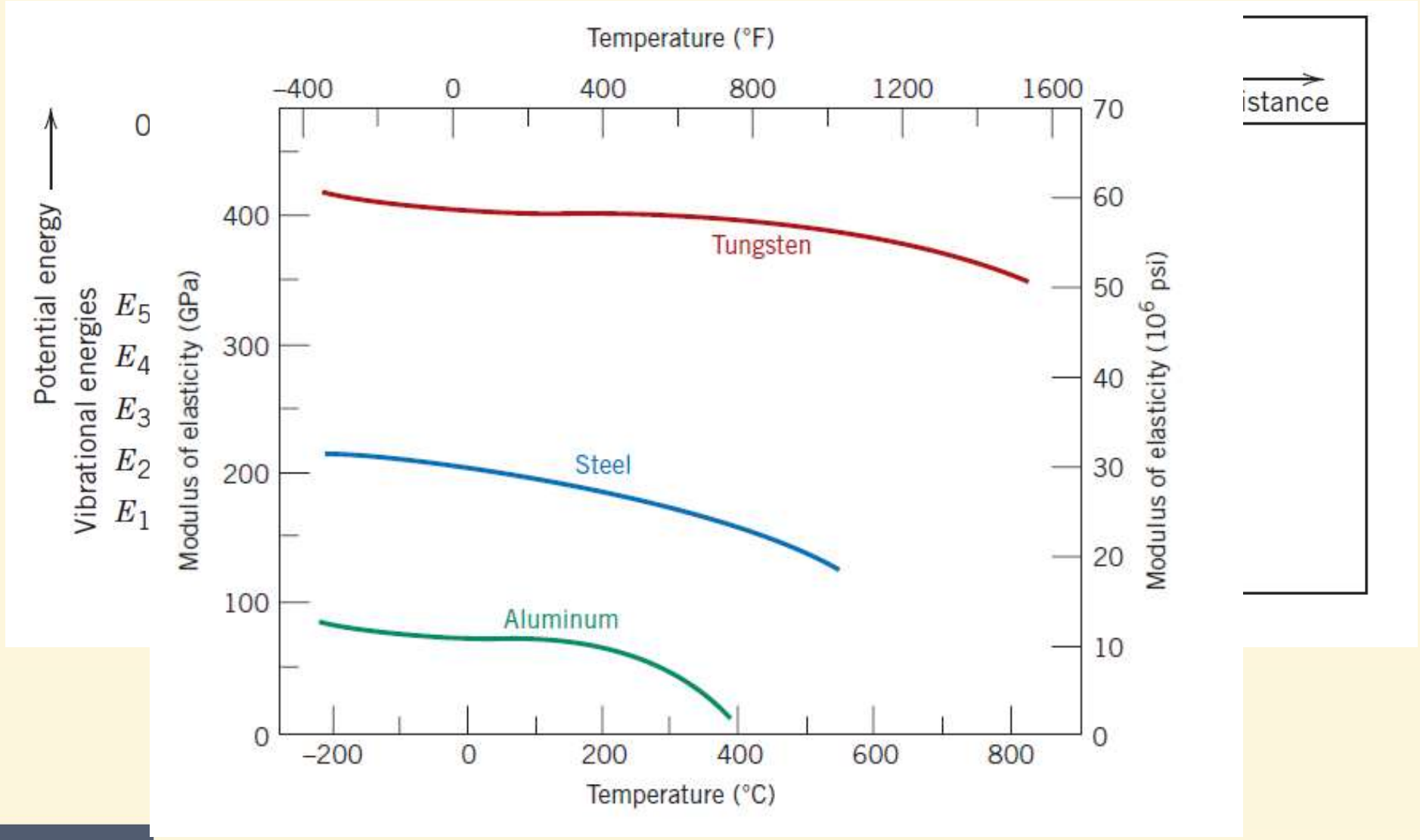
Table 19.1 Tabulation of the Thermal Properties for a Variety of Materials

Material	c_p *(J/kg-K)*[a]	α_l *[(°C)$^{-1}$ × 10^{-6}]*[b]	k *(W/m-K)*[c]	L *[Ω-W/(K)^2 × 10*
		Metals		
Aluminum	900	23.6	247	2.20
Copper	386	17.0	398	2.25
Gold	128	14.2	315	2.50
Iron	448	11.8	80	2.71
Nickel	443	13.3	90	2.08
Silver	235	19.7	428	2.13
Tungsten	138	4.5	178	3.20
1025 Steel	486	12.0	51.9	—
316 Stainless steel	502	16.0	15.9	—
Brass (70Cu–30Zn)	375	20.0	120	—
Kovar (54Fe–29Ni–17Co)	460	5.1	17	2.80
Invar (64Fe–36Ni)	500	1.6	10	2.75
Super Invar (63Fe–32Ni–5Co)	500	0.72	10	2.68
		Ceramics		
Alumina (Al_2O_3)	775	7.6	39	—
Magnesia (MgO)	940	13.5[d]	37.7	—
Spinel ($MgAl_2O_4$)	790	7.6[d]	15.0[e]	—
Fused silica (SiO_2)	740	0.4	1.4	—
Soda–lime glass	840	9.0	1.7	—
Borosilicate (Pyrex™) glass	850	3.3	1.4	—
		Polymers		
Polyethylene (high density)	1850	106–198	0.46–0.50	—
Polypropylene	1925	145–180	0.12	—
Polystyrene	1170	90–150	0.13	—
Polytetrafluoroethylene (Teflon™)	1050	126–216	0.25	—
Phenol-formaldehyde, phenolic	1590–1760	122	0.15	—
Nylon 6,6	1670	144	0.24	—
Polyisoprene	—	220	0.14	—

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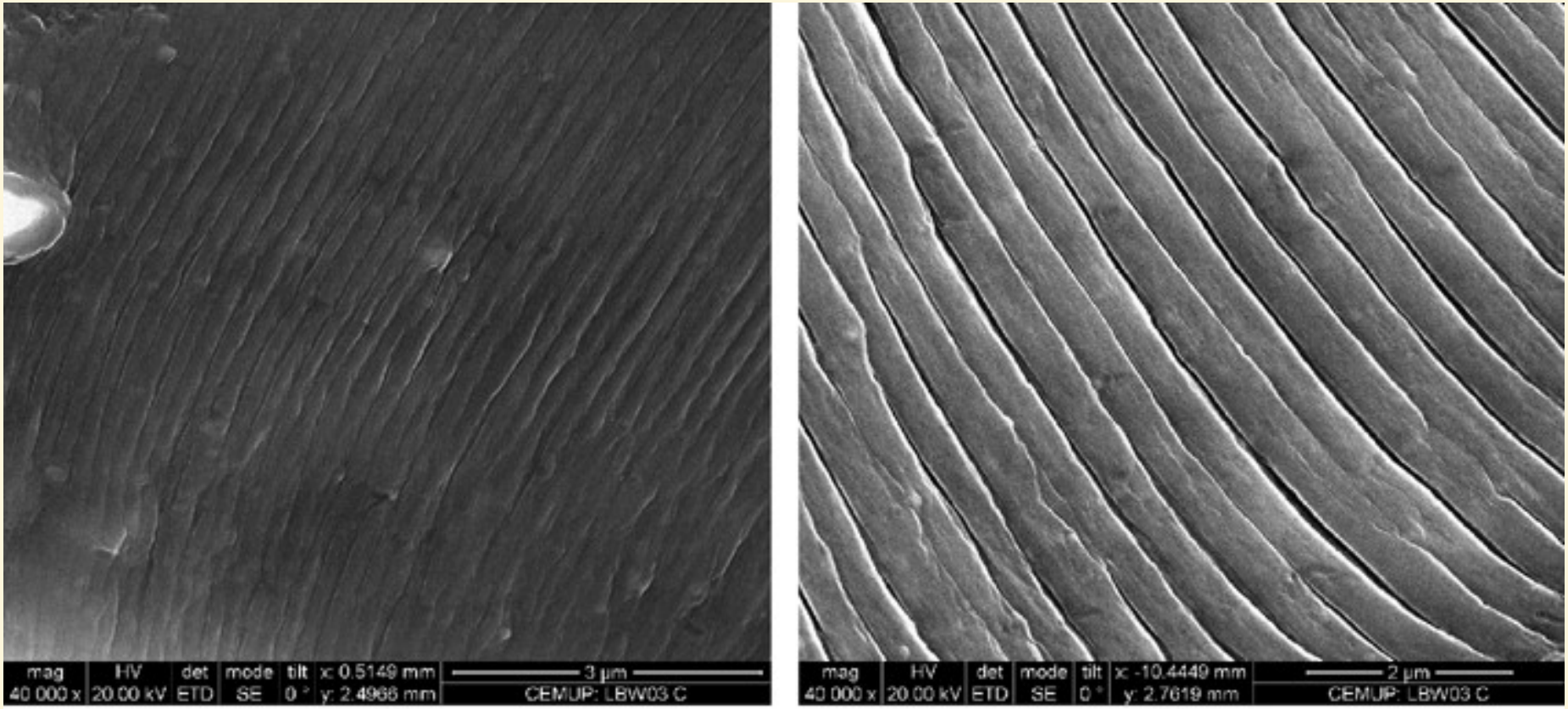
Thermal Properties

- Thermal Expansion



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Fracture/Fatigue/Creep



(a) **(b)**

Notes: (a) Fractograph 6 of sample C, $x = 13.03$ mm, $y = 1.09$ mm, $s = 0.19\ \mu$m;
(b) fractograph 20 of sample C, $x = 24.00$ mm, $y = 1.35$ mm, $s = 0.42\ \mu$m

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Density and Corrosion

Density: determined by the mass and size of the atoms and how they are packed together

1. Ceramics and polymers – light elements (C, O and Si) SiC 2.5 Mg/m^3, Nylon 1.0 Mg/m^3
2. Metals heavier elements which are tightly packed together – high density Iron 7.8 Mg/m^3

Corrosion: A chemical reaction between a material and its environment

1. Problem with metals (apart from noble metals such as Gold)
2. Ceramic: more stable especially for oxides
3. Polymers can have very good corrosion resistance but depends on chemistry of the polymers

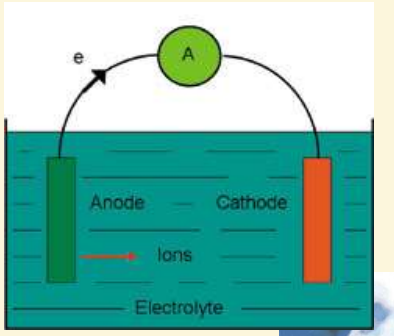
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Corrosion: Anodic and Cathodic reactions

Red. ↔Ox + e⁻	E^0 [V]
$Li \leftrightarrow Li^+ + e^-$	-3.045
$K \leftrightarrow K^+ + e^-$	-2.925
$Ca \leftrightarrow Ca^{2+} + 2e^-$	-2.866
$Na \leftrightarrow Na^+ + e^-$	-2.714
$Mg \leftrightarrow Mg^{2+} + 2e^-$	-2.363
$Al \leftrightarrow Al^{3+} + 3e^-$	-1.662
$Mn \leftrightarrow Mn^{2+} + 2e^-$	-1.180
$Zn \leftrightarrow Zn^{2+} + 2e^-$	-0.7627
$Cr \leftrightarrow Cr^{3+} + 3e^-$	-0.744
$Cd \leftrightarrow Cd^{2+} + 2e^-$	-0.4029
$Fe \leftrightarrow Fe^{2+} + 2e^-$	-0.4002
$Co \leftrightarrow Co^{2+} + 2e^-$	-0.277
$Ni \leftrightarrow Ni^{2+} + 2e^-$	-0.250
$Pb \leftrightarrow Pb^{2+} + 2e^-$	-0.126
$H_2 \leftrightarrow 2H^+ + 2e^-$	±0.0000
$CH_4 \leftrightarrow C + 4H^+ + 4e^-$	+0.1316
$Cu^+ \leftrightarrow Cu^{2+} + e^-$	+0.153
$Cu \leftrightarrow Cu^{2+} + 2e^-$	+0.337
$2OH^- \leftrightarrow ½ O_2 + H_2O + 2e^-$	+0.401
$I^- \leftrightarrow ½ I_2 + e^-$	+0.5355
$Fe^{2+} \leftrightarrow Fe^{3+} + e^-$	+0.771
$Ag \leftrightarrow Ag^+ + e^-$	+0.7991
$Hg \leftrightarrow Hg^{2+} + 2e^-$	+0.854
$Hg_2^{2+} \leftrightarrow 2Hg^{2+} + 2e^-$	+0.9075
$Pd \leftrightarrow Pd^{2+} + 2e^-$	+0.987
$Br^- \leftrightarrow ½ Br_2 + e^-$	+1.0652
$Pt \leftrightarrow Pt^{2+} + 2e^-$	~+1.2
$Cl^- \leftrightarrow ½ Cl_2 + e^-$	+1.3595
$Au \leftrightarrow Au^{3+} + 3e^-$	+1.402
$Au \leftrightarrow Au^{3+} + 3e^-$	+1.498
$Pb^{2+} \leftrightarrow Pb^{4+} + 4e^-$	+1.80
$Ag \leftrightarrow Ag^{2+} + e$	+1.980
$F^- \leftrightarrow ½ F_2 + e^-$	+2.87
$HF\ (aq) \leftrightarrow ½ F_2 + H^+ + e^-$	+3.06

- Power of reduction increases
- Oxidability increases
- Metals become ignoble

- Metals become noble
- Power of oxidation increase
- Reductibility increases



$$E_{red} = E^{\ominus}_{red} - \frac{RT}{zF} \ln \frac{a_{Red}}{a_{Ox}}$$







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Ellingham Diagram

